

0 Single-variable warm-up (test Q1–6)

- **Limits.** L'Hôpital for $\frac{0}{0}, \frac{\infty}{\infty}$: $\lim_{x \rightarrow \infty} \frac{f}{g} = \lim_{x \rightarrow \infty} \frac{f'}{g'}$.
 Rewrite $0 \cdot \infty$ as $f/(1/g)$; for $1^\infty, 0^0, \infty^0$ take
 ln first. **Faster: Maclaurin series** near 0:
 $e^x = 1 + x + x^2/2 + x^3/6$ $\sin x = x - x^3/6$
 $\cos x = 1 - x^2/2 + x^4/24$ $\tan x = x + x^3/3$
 $\ln(1+x) = x - x^2/2 + x^3/3$
 $\arctan x = x - x^3/3$
 $(1+x)^k \approx 1 + kx + k(k-1)/2 \cdot x^2$

PY Q1 $\lim_{x \rightarrow 0} \frac{e^{2x} - \cos x - 2 \sin x}{x^2}$: numerator = $(1 + 2x + 2x^2) - (1 - x^2/2) - 2x + O(x^3) = 5/2 \cdot x^2 \Rightarrow$
 limit = $5/2$.

- **Implicit / related rates.** $F(x, y) = 0 \Rightarrow \frac{dy}{dx} = -F_x/F_y$. Horizontal tangent: solve $F_x = 0$ with $F = 0$, **check $F_y \neq 0$** vertical: $F_y = 0$. Related rates: d/dt both sides, **then** substitute.

PY Q2–3 $x^2 - 7y^2 + y^3 = 6xy$: $F_x = 2x - 6y = 0 \Rightarrow x = 3y \Rightarrow y^3 = 16y^2$; $(0, 0)$ is singular ($F_y = 0$), so only $(48, 16)$. $2x^3 + 3y^2 + xy = 6$ at $(1, 1)$, $x' = 4$: $6x^2x' + 6yy' + x'y + xy' = 0 \Rightarrow y' = -4$.

- **FTC.** $\frac{d}{dx} \int_a^{b(x)} f(t) dt = f(b(x))b'(x) - f(a(x))a'(x)$.

PY Q5 $f = \int_{-3x}^x (t+1) dt$: $f'(1) = (1+1)(3) - (-\frac{1}{3} + 1)(2) = \frac{10}{3}$.

- **Integrals.** u -sub (change the limits!); parts $\int u dv = uv - \int v du$ (LIATE); partial fractions. $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$, e.g. $\int_0^\infty x e^{-x/2} dx = 1/(1/2)^2 = 4$.

PY Q6 $\int_0^{\ln 3} e^x(1+e^x)^2 dx$, $u = 1 + e^x$: $\int_2^4 u^2 du = (64 - 8)/3 = 56/3$.

- **Improper.** Replace the bad limit by $\lim_{b \rightarrow \infty}$; split at interior singularities. $\int_1^\infty x^{-p}$ converges iff $p > 1$; $\int_0^1 x^{-p}$ iff $p < 1$.
 ► **Polar (10.3–10.4).** $x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$, $\tan \theta = y/x$. $r = 2a \cos \theta$: circle $(x-a)^2 + y^2 = a^2$; $r = a(1 - \cos \theta)$: cardioid.
 ► **Applications.** Disk $\pi \int R^2$; washer $\pi \int (R^2 - r^2)$; shell $2\pi \int (\text{radius})(\text{height})$; arc length $\int_a^b \sqrt{1 + (y')^2} dx$; $f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$.

11.3 Vectors & the dot product

- $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$; unit vector $\mathbf{v}/|\mathbf{v}|$; vector of length L along $\mathbf{v}$: $L\mathbf{v}/|\mathbf{v}|$. Point dividing PQ in ratio $r : s$: $(sP + rQ)/(r + s)$.
 ► $\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = |\mathbf{u}||\mathbf{v}| \cos \theta$, $\theta = \cos^{-1} \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} \in [0, \pi]$. $\mathbf{u} \perp \mathbf{v} \Leftrightarrow \mathbf{u} \cdot \mathbf{v} = 0$; acute $\Leftrightarrow > 0$; obtuse $\Leftrightarrow < 0$. (PY Q7: $2c - 15 + 6 = 0 \Rightarrow c = 9/2$.)
 ► $\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}$, scalar component $(\mathbf{u} \cdot \mathbf{v})/|\mathbf{v}|$.

Decompose $\mathbf{u} = \frac{\text{proj}_{\mathbf{v}} \mathbf{u}}{\|\mathbf{v}\|} + \frac{(\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u})}{\|\mathbf{v}\|}$. Work $W = \mathbf{F} \cdot \overrightarrow{PQ}$.

PY Q8 $\mathbf{Q} \perp \mathbf{V}$, $\mathbf{U} - \mathbf{Q} \parallel \mathbf{V} \Rightarrow \mathbf{U} - \mathbf{Q} = \text{proj}_{\mathbf{V}} \mathbf{U}$. $\mathbf{U} = \langle 4, 3, 4 \rangle$, $\mathbf{V} = \langle 1, 2, 2 \rangle$: $(\mathbf{U} \cdot \mathbf{V})/|\mathbf{V}|^2 = 18/9 = 2$, so $\mathbf{Q} = \langle 4, 3, 4 \rangle - 2\langle 1, 2, 2 \rangle = 2\mathbf{i} - \mathbf{j}$.

11.4 Cross product & determinants

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$

- $\mathbf{u} \times \mathbf{v} \perp \mathbf{u}, \mathbf{v}$; $\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}$; $\mathbf{u} \parallel \mathbf{v} \Leftrightarrow \mathbf{u} \times \mathbf{v} = \mathbf{0}$; $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$.
 ► $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin \theta$ = **parallelogram area**; tri-angle = $\frac{1}{2} |\mathbf{AB} \times \mathbf{AC}|$.

- $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \det[\mathbf{u}; \mathbf{v}; \mathbf{w}] = \pm$ **parallelepiped volume**; tetrahedron $\frac{1}{6} |\det|$. $\det = 0 \Leftrightarrow$ coplanar.
 ► Cofactor expansion along any row/column with signs $(-1)^{i+j}$ (checkerboard $+-+$). Row swap flips the sign; $\det(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \det(\mathbf{w}, \mathbf{u}, \mathbf{v})$ (cyclic).

11.5 Lines & planes — recipes

Line through P_0 along $\mathbf{v}$: $\mathbf{r} = P_0 + t\mathbf{v}$, i.e. $x = x_0 + tv_1, \dots$; segment $P \rightarrow Q$: $P + t(Q - P)$, $0 \leq t \leq 1$. **Plane** through P_0 with normal $\mathbf{n} = \langle a, b, c \rangle$: $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$. Normal = coefficients.

Finding the normal $\mathbf{n}$

- 3 points P, Q, R : $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$.
 ► point A and line $(P_0, \mathbf{v})$: $\mathbf{n} = \mathbf{v} \times \overrightarrow{P_0 A}$.
 ► contains A, B , $\perp$ plane with normal $\mathbf{m}$: $\mathbf{n} = \mathbf{AB} \times \mathbf{m}$.
 ► contains the line $\Pi_1 \cap \Pi_2$ and point A : **pen-cil** $(a_1 x + \dots - d_1) + \lambda(a_2 x + \dots - d_2) = 0$, fix λ by A .
 ► line $\perp$ to two lines: direction $\mathbf{v}_1 \times \mathbf{v}_2$. Line $\Pi_1 \cap \Pi_2$: direction $\mathbf{n}_1 \times \mathbf{n}_2$, point: set $z = 0$, solve.

PY Q10 $L : (0, 0, 1) + t(1, 2, 0)$, $A = (1, 1, 3)$: $\overrightarrow{P_0 A} = \langle 1, 1, 2 \rangle$, $\mathbf{n} = \langle 1, 2, 0 \rangle \times \langle 1, 1, 2 \rangle = \langle 4, -2, -1 \rangle \Rightarrow 4x - 2y - z = -1$, i.e. $z = 4x - 2y + 1$.

- **Line $\cap$ plane:** substitute $x(t), y(t), z(t)$ into the plane, solve for t .

PY Q11 $(1+t, 1+2t, 1+t)$ in $2x - y + 3z = 10$: $4 + 3t = 10$, $t = 2 \Rightarrow (3, 5, 3)$.

- Line $\parallel$ plane $\Leftrightarrow \mathbf{v} \cdot \mathbf{n} = 0$; line $\perp$ plane $\Leftrightarrow \mathbf{v} \parallel \mathbf{n}$. Planes parallel $\Leftrightarrow \mathbf{n}_1 \parallel \mathbf{n}_2$; angle between planes = angle between normals.
 ► Angle between line and plane: $\sin \varphi = \frac{|\mathbf{v} \cdot \mathbf{n}|}{|\mathbf{v}||\mathbf{n}|}$. Foot of perpendicular from S to line $P_0 + t\mathbf{v}$: $t = (\overrightarrow{P_0 S} \cdot \mathbf{v})/|\mathbf{v}|^2$. Collinear $A, B, C \Leftrightarrow \overrightarrow{AB} \times \overrightarrow{AC} = \mathbf{0}$.
 ► Two lines: parallel ($\mathbf{v}_1 \parallel \mathbf{v}_2$), intersecting (solve $P_1 + t\mathbf{v}_1 = P_2 + s\mathbf{v}_2$ consistently), else **skew**.
 ► Distance S to plane: $\frac{|\mathbf{ax} + \mathbf{by} + \mathbf{cz} - d|}{\sqrt{a^2 + b^2 + c^2}}$; S to line: $\frac{|\overrightarrow{P_0 S} \times \mathbf{v}|}{|\mathbf{v}|}$; parallel planes: $|d_1 - d_2|/|\mathbf{n}|$; skew lines: $\frac{|\overrightarrow{P_1 P_2} \cdot (\mathbf{v}_1 \times \mathbf{v}_2)|}{|\mathbf{v}_1 \times \mathbf{v}_2|}$. E.g. $(2, -1, 1)$ to $x + y - z = 2$: $|2 - 1 - 1|/\sqrt{3} = 2/\sqrt{3}$.
 ► Answer " $ax + by + cz = 1$ ": divide by the constant d (fractions!). Intercept form $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

PY Q9 $(1, 2, -1), (3, 1, 0), (-1, 0, 2)$: $\mathbf{n} = \langle 2, -1, 1 \rangle \times \langle -2, -2, 3 \rangle = \langle -1, -8, -6 \rangle$, so $x + 8y + 6z = 11 \Rightarrow \frac{1}{11}x + \frac{8}{11}y + \frac{6}{11}z = 1$.

12.1–12.2 Vector functions & motion

- $\mathbf{r}(t) = \langle f, g, h \rangle$; velocity $\mathbf{v} = \mathbf{r}'$, **speed** $|\mathbf{v}|$, acceleration $\mathbf{a} = \mathbf{r}''$, unit tangent $\mathbf{T} = \mathbf{v}/|\mathbf{v}|$. Limits, derivatives, integrals: **component-wise**.
 ► $(\mathbf{u} \cdot \mathbf{v})' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$; $(\mathbf{u} \times \mathbf{v})' = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$ (keep order); $(\mathbf{u}(f(t)))' = f'(t)\mathbf{u}'(f(t))$.
 ► $|\mathbf{r}|$ constant $\Leftrightarrow \mathbf{r} \cdot \mathbf{r}' = 0$ (motion on a sphere). Constant speed $\Rightarrow \mathbf{v} \perp \mathbf{a}$.
 ► Angle between curves at a common point = angle between their tangent vectors. Max/min speed: extremise $|\mathbf{v}|^2$.
 ► **Tangent line** at t_0 : $\mathbf{r}(t_0) + s\mathbf{r}'(t_0)$. **Initial values:** $\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^t \mathbf{v}(\tau) d\tau$.

PY Q12 $\mathbf{r} = \langle 3 - t^2, t^3, 4t - 1 \rangle$, $t = 1$: $P = (2, 1, 3)$, $\mathbf{r}' = \langle -2, 3, 4 \rangle$. $L : (2 - 2s, 1 + 3s, 3 + 4s)$; yz -plane $x = 0 \Rightarrow s = 1 \Rightarrow (0, 4, 7)$.

e.g. $\mathbf{v} = \langle 6t^2 + 4t, 8t^3 + 2t, 1/(1+t^2) \rangle$, $\mathbf{r}(0) = \langle 1, 2, \pi \rangle$; $\mathbf{r}(1) = \mathbf{r}(0) + \int_0^1 \mathbf{v} = \langle 1 + 4, 2 + 3, \pi + \pi/4 \rangle = \langle 5, 5, 5\pi/4 \rangle$.

- **Collide:** same t in $\mathbf{r}_1(t) = \mathbf{r}_2(t)$. **Paths meet:** $\mathbf{r}_1(t) = \mathbf{r}_2(s)$ with independent t, s .

► Projectile (v_0 , angle α , origin): $\mathbf{r} = (v_0 \cos \alpha)t\mathbf{i} + (v_0 \sin \alpha)t - \frac{1}{2}gt^2\mathbf{j}$; $y_{\max} = \frac{(v_0 \sin \alpha)^2}{2g}$, flight time $\frac{2v_0 \sin \alpha}{g}$, range $\frac{v_0^2 \sin 2\alpha}{g}$.

12.3 Arc length

$L = \int_a^b |\mathbf{v}(t)| dt = \int_a^b \sqrt{x'^2 + y'^2 + z'^2} dt$; arc-length parameter $s(t) = \int_{t_0}^t |\mathbf{v}(\tau)| d\tau$ ($|\mathbf{v}| = 1 \Leftrightarrow$ parametrised by arc length).

Simplify $|\mathbf{v}|^2$ before rooting: $\sin^2 + \cos^2 = 1$; $1 - \cos t = 2 \sin^2(t/2)$; $1 + \cos t = 2 \cos^2(t/2)$; perfect squares. Keep $|\cdot|$: $\sqrt{\sin^2(t/2)} = |\sin(t/2)|$ — split the interval where the sign changes.

e.g. $\langle 2t - 2 \sin t, 2 - \sqrt{2} \cos t, \sqrt{2} \cos t \rangle$, $0 \leq t \leq 4\pi$: $|\mathbf{v}|^2 = (2 - 2 \cos t)^2 + 4 \sin^2 t = 8(1 - \cos t) = 16 \sin^2(t/2) \Rightarrow L = \int_0^{4\pi} 4|\sin(t/2)| dt = 2 \cdot 16 = 32$.

Helix $\langle a \cos t, a \sin t, bt \rangle$: $|\mathbf{v}| = \sqrt{a^2 + b^2}$. $\int \sqrt{t^2 + a^2} dt = \frac{t}{2} \sqrt{t^2 + a^2} + \frac{a^2}{2} \ln(t + \sqrt{t^2 + a^2})$.

13.1–13.2 Functions, domains, limits

- **Domain:** $\sqrt{\cdot} \geq 0$, $\ln(\cdot) > 0$, denominators $\neq 0$, $\sin^{-1}$ argument in $[-1, 1]$.
 ► Level curves $f(x, y) = k$; level surfaces $f(x, y, z) = k$. $x^2 + y^2$: circles / paraboloid; $\sqrt{x^2 + y^2}$: circles / cone; $(x + y)/(x - y) = k$: lines through O .
 ► **Limits:** continuous $\Rightarrow$ substitute. Else simplify — factor, conjugate ($\frac{\sqrt{u}-2}{u-4} = \frac{1}{\sqrt{u}+2}$), sub $u = 2x - y$.
 ► **Exists:** squeeze $|f - L| \leq g \rightarrow 0$; polar $x = r \cos \theta$, $y = r \sin \theta$ — if bound $\rightarrow 0$ as $r \rightarrow 0$ for **all** θ , limit is 0 (e.g. $\frac{x^2 y}{x^2 + y^2} = r \cos^2 \theta \sin \theta$).
 ► **DNE (two-path test):** different values along $y = mx$, $y = kx^2$, axes. E.g. $\frac{xy}{x^2 + y^2} \rightarrow \frac{m}{1+m^2}$; $\frac{y}{x^2} \rightarrow k$ along $y = kx^2$.
 ► Continuous at $(a, b) \Leftrightarrow \lim = f(a, b)$ (limit exists **and** equals value).

13.3 Partial derivatives

- f_x : differentiate in x , **all others constant**. $f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y \partial x}$.
 ► **Clairaut:** $f_{xy} = f_{yx}$ (continuous 2nd partials). So $f_x = P$, $f_y = Q$ has a solution only if $P_y = Q_x$ — use to find unknown constants.
 ► **Implicit:** $F(x, y, z) = 0 \Rightarrow \frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$, $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$.

Rebuild f from f_x, f_y : $f = \int f_x dx + g(y)$; differentiate in y , match $f_y \Rightarrow g'(y)$; fix the constant from a given value.

PY Q13 $f_x = 2x + y^2$, $f_y = 2xy + 3y^2$: $f = x^2 + xy^2 + g(y)$, $g' = 3y^2 \Rightarrow f = x^2 + xy^2 + y^3 + C$. $f(2, 1) = 7 + C = \frac{11}{2} \Rightarrow C = -\frac{3}{2}$; $f(1, 0) = -\frac{1}{2}$.

- Laplace $f_{xx} + f_{yy} = 0$; wave $f_{tt} = c^2 f_{xx}$. f_x, f_y continuous $\Rightarrow$ differentiable $\Rightarrow$ continuous.

13.4 The Chain Rule

- $w = f(x, y, z)$, all functions of t : $\frac{dw}{dt} = f_x x' + f_y y' + f_z z' = \nabla f \cdot \mathbf{r}'(t)$.
- x, y functions of (u, v) : $\frac{\partial w}{\partial u} = f_x x_u + f_y y_u$ — **sum over every path** in the tree diagram.
- $w = f(u)$, $u = g(x, y)$: $w_x = f'(u)u_x$. Implicit $F(x, y) = 0$: $\frac{dy}{dx} = -\frac{F_x}{F_y}$.
- Polar: $w_r = f_x \cos \theta + f_y \sin \theta$, $w_\theta = -f_x r \sin \theta + f_y r \cos \theta$; $f_x^2 + f_y^2 = w_r^2 + w_\theta^2 / r^2$.
- PY Q16** $w = u^2 + 2uv$, $u = xy$, $v = -y^2$: $w_x = w_u u_x + w_v v_x = (2u + 2v)y + 0 = 2xy^2 - 2y^3$.
- PY Q17** $\nabla f(1, 1, 1) = \langle a, b, c \rangle$, $\langle x', y', z' \rangle = \langle 5, 5, 5 \rangle \Rightarrow \frac{df}{dt} = 5(a + b + c)$.

13.5 Gradient & directional derivative

$\nabla f = \langle f_x, f_y, f_z \rangle$. $D_{\mathbf{u}}f(P) = \nabla f(P) \cdot \mathbf{u}$ with $\mathbf{u}$ a **unit** vector ($\mathbf{u} = \mathbf{v}/|\mathbf{v}|$); $D_{\mathbf{u}}f = |\nabla f| \cos \theta$.

- fastest increase: along ∇f , rate $|\nabla f|$; fastest decrease: $-\nabla f$, rate $|\nabla f|$;
- no change: $\mathbf{u} \perp \nabla f$ (2-D: $\mathbf{u} = \pm \langle -f_y, f_x \rangle / |\nabla f|$);
- attainable rates: $|\nabla f| \leq D_{\mathbf{u}}f \leq |\nabla f|$ ("is there a direction with rate 14?").
- ∇f is $\perp$ to the level curve/surface through P ; tangent line to level curve: $f_x(x - x_0) + f_y(y - y_0) = 0$.
- Given $D_{\mathbf{u}_1}f, D_{\mathbf{u}_2}f$: solve the 2×2 system for f_x, f_y . Along a path: $df/dt = \nabla f \cdot \mathbf{v}$; per unit distance: $\nabla f \cdot \mathbf{T}$.

e.g. $D_{\mathbf{u}}f = 2\sqrt{2}$ along $\mathbf{i} + \mathbf{j}$ and -3 along $-2\mathbf{j}$: $f_y = 3/2$, $(f_x + f_y)/\sqrt{2} = 2\sqrt{2} \Rightarrow f_x = 5/2$; along $-\mathbf{i} - 2\mathbf{j}$: $(-5/2 - 3)/\sqrt{5} = -11/(2\sqrt{5})$.

PY Q18 $f = a \ln x + xy^2$: $\nabla f(1, 2) = \langle a + 4, 4 \rangle$, $\mathbf{u} = \langle 4, -3 \rangle/5$: $(4(a + 4) - 12)/5 = 2 \Rightarrow a = \frac{3}{2}$.

PY Q14 $\nabla(x^2y + y^2z + z^2x) = \langle 2xy + z^2, x^2 + 2yz, y^2 + 2zx \rangle$; at $(1, 2, 3)$: $\langle 13, 13, 10 \rangle$.

13.6 Tangent planes, linearization, differentials

- Level surface** $F = c$ at P_0 : $\nabla F(P_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$; normal line $P_0 + t\nabla F(P_0)$.
- Graph** $z = f(x, y)$: $z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ (= level surface $F = f - z$).
- Tangent line to the curve $F = c \cap G = k$: direction $\nabla F \times \nabla G$.

e.g. $x^3 + 2xy + yz = 7 \cap 3x^2 - yz = 1$ at $(1, 2, 1)$: $\nabla F = \langle 7, 3, 2 \rangle$, $\nabla G = \langle 6, -1, -2 \rangle$, $\nabla F \times \nabla G = \langle -4, 26, -25 \rangle \Rightarrow x = 1 - 4t, y = 2 + 26t, z = 1 - 25t$.

PY Q15 $x^2 + y^3 + z^4 = 8$ at $(3, 2, 1)$: $\nabla F = \langle 6, 12, 4 \rangle \Rightarrow 6x + 12y + 4z = 46 \Rightarrow \frac{3}{23}x + \frac{6}{23}y + \frac{2}{23}z = 1$.

- Linearization** $L(x, y) = f(a, b) + f_x(x - a) + f_y(y - b)$.
- Differential** $df = f_x dx + f_y dy + f_z dz$, $\Delta f \approx df$. Moving distance ds along $\mathbf{u}$: $df \approx (\nabla f \cdot \mathbf{u})ds$.
- Relative error: $\frac{df}{f} = d(\ln f)$; for $f = kx^a y^b$: $\frac{df}{f} = a \frac{dx}{x} + b \frac{dy}{y}$. Implicit: take $d(\cdot)$ of both sides, solve for dw .

e.g. $V = \pi r^2 h$ with r off by 2%, h by 1%: $\frac{dV}{V} = 2 \frac{dr}{r} + \frac{dh}{h} \Rightarrow$ at most 5%.

13.7 Extreme values & saddle points

Critical point: $f_x = f_y = 0$ (or undefined). At it, $D = f_{xx}f_{yy} - f_{xy}^2$:

$D > 0$	$D > 0$	$D < 0$	$D = 0$
$f_{xx} < 0$	$f_{xx} > 0$		
local max	local min	saddle	no info

You never need f itself — only its 2nd partials at the point. If $D = 0$: test $f - f(a, b)$ along lines/curves — at $(0, 0)$, x^2y^2 is a min, $1 - x^2y^2$ a max, xy^2 neither.

PY Q21 $f_x = 2x + \cos y$, $f_y = -x \sin y + 2y - \pi$; at $(0, \pi/2)$: $f_{xx} = 2$, $f_{yy} = -x \cos y + 2 = 2$, $f_{xy} = -\sin y = -1 \Rightarrow D = 3 > 0 \Rightarrow$ **local min**.

PY Q20 $f = e^x(x^2 - y^2)$ at $(0, 0)$: $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0 \Rightarrow D = -4 \Rightarrow$ **saddle**.

- Solving $\nabla f = 0$** : factor and split cases ($y(2 - 2x - y) = 0$); substitute the simpler equation into the other; add/subtract; symmetry $\Rightarrow$ try $x = y$.

PY Q19 $f = \frac{1}{4}x^4 - \frac{1}{2}x^2 - xy + \frac{1}{2}y^2$: $f_y = y - x = 0$, $f_x = x^3 - x - y = x^3 - 2x = 0 \Rightarrow (0, 0), (\pm\sqrt{2}, \pm\sqrt{2})$.

- Absolute extrema on closed bounded R** : (1) interior critical points; (2) each boundary piece as a 1-variable function (circle: $x = r \cos t$, $y = r \sin t$); (3) corners. Compare all values.

e.g. $x^2 - 2xy + 2y^2 - 2y$ on $[-1, 2] \times [0, 2]$: interior $(1, 1) \rightarrow -1$; edge $x = 2$: $2y^2 - 6y + 4$, vertex $y = 3/2 \rightarrow -1/2$; corners $1, 9, 4, 0 \Rightarrow$ max 9 at $(-1, 2)$, min -1 at $(1, 1)$.

- Distance problems**: minimise d^2 , eliminate a variable via the surface (e.g. $y^2 = 4 + xz \Rightarrow d^2 = x^2 + xz + z^2 + 4$). Box with volume V : $z = V/(xy)$.

LSQ Least squares (13.7)

Fit $y = ag(x) + bh(x)$: minimise $S = \sum (ag_i + bh_i - y_i)^2$; $S_a = S_b = 0$ give the **normal equations**

$$a \sum g_i^2 + b \sum g_i h_i = \sum g_i y_i$$

$$a \sum g_i h_i + b \sum h_i^2 = \sum h_i y_i$$

Line $y = mx + b$: $m \sum x^2 + b \sum x = \sum xy$, $m \sum x + nb = \sum y$. Tabulate $x, g, h, y, g^2, gy, \dots$

e.g. Line through $(-1, 1), (1, 2), (3, 2)$: $\sum x = 3$, $\sum x^2 = 11$, $\sum y = 5$, $\sum xy = 7 \Rightarrow 11m + 3b = 7$, $3m + 3b = 5 \Rightarrow y = \frac{1}{4}x + \frac{17}{12}$.

PY Q22 $y = ax^3 + b$ through $(-1, 1), (0, -7), (1, 7)$: $g = x^3 \in \{-1, 0, 1\}$, $h = 1$: $\sum g^2 = 2$, $\sum g = 0$, $\sum gy = 6$, $\sum y = 1 \Rightarrow 2a = 6$, $3b = 1 \Rightarrow y = 3x^3 + \frac{1}{3}$.

13.8 Lagrange multipliers

Extremise f subject to $g = c$: solve $\nabla f = \lambda \nabla g$, $g = c$. Two constraints: $\nabla f = \lambda \nabla g + \mu \nabla h$. Then **evaluate f at every solution** — largest = max, smallest = min (guaranteed if the constraint set is closed & bounded; on $xy = 16$, $x + y$ has no max).

Solving tricks

- eliminate λ : $f_x/g_x = f_y/g_y$ — check $g_x = 0$, $\lambda = 0$, variable = 0 cases separately;
- $f = xyz$: multiply the equations by $x, y, z \Rightarrow \lambda x g_x = \lambda y g_y = \lambda z g_z$;
- region with an inequality ($x \leq 2$, disk): interior critical points + Lagrange on the boundary + corners.

e.g. xy on $(x - 2)^2 + y^2 = 4$: $y^2 = x^2 - 2x \Rightarrow x = 0$ or 3, max $3\sqrt{3}$ at $(3, \sqrt{3})$. Add $x \leq 2$: $x = 3$ is cut off, check the new edge $x = 2$: $y = \pm 2 \Rightarrow$ max 4 at $(2, 2)$.

PY Q23 $2x + y$ on $2x^2 + y^2 = 9$: $2 = 4\lambda x$, $1 = 2\lambda y \Rightarrow x = y \Rightarrow 3x^2 = 9$, max = $3\sqrt{3}$.

PY Q24 xyz on $x^2 + y^2 + z^2 + 2xz = 18$: constraint = $(x + z)^2 + y^2$ is symmetric in $x, z \Rightarrow x = z$; then x^2y on $4x^2 + y^2 = 18 \Rightarrow y^2 = 6$, $x^2 = 3$: $(\sqrt{3}, \sqrt{6}, \sqrt{3})$.

e.g. Open box, $V = xyz = 6$, cost $20xz + 10yz + 30xy$: substitute $z = 6/(xy)$, or Lagrange $\Rightarrow (x, y, z) = (1, 2, 3)$, minimum \$180 (each cost term equals \$60 at the optimum).

- Shortcuts: max of $ax + by + cz$ on $x^2 + y^2 + z^2 = R^2$ is $R\sqrt{a^2 + b^2 + c^2}$ at $R\langle a, b, c \rangle / |\langle a, b, c \rangle|$; max of $ax + by$ on $px^2 + qy^2 = k$ is $\sqrt{k(a^2/p + b^2/q)}$.
- First-octant box, corner on $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$: $V_{\max} = (abc)/(3\sqrt{3})$ at $x = a/\sqrt{3}, \dots$. Fixed sum S : max product $(S/3)^3$, min sum of squares when all equal.
- Nearest point of plane $\mathbf{n} \cdot \mathbf{x} = d$ to P : $P + t\mathbf{n}$, $t = (d - \mathbf{n} \cdot P)/|\mathbf{n}|^2$. Nearest/farthest point of sphere radius R (centre O) to P : $\pm RP/|P|$.

13.9–13.10 Taylor & constrained variables

- 2nd-order Taylor at (a, b) , with $h = x - a$, $k = y - b$:

$$f \approx f(a, b) + f_x h + f_y k$$

$$+ \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)$$

- At $(0, 0)$: **multiply 1-variable series, truncate at degree 2** ($u = x + 4y$):

$$e^u \cos x \approx (1 + u + u^2/2)(1 - x^2/2)$$

$$\approx 1 + x + 4y + 4xy + 8y^2$$

- $(\frac{\partial w}{\partial x})_y$: x, y independent, the rest dependent. Eliminate the dependent ones via the constraint, then differentiate.

e.g. $w = x^2 + y^2 + z^2$, $z = x^2 + y^2$. $(\frac{\partial w}{\partial y})_z$: $x^2 = z - y^2 \Rightarrow w = z + z^2 \Rightarrow 0$. $(\frac{\partial w}{\partial y})_x$: $w = x^2 + y^2 + (x^2 + y^2)^2 \Rightarrow 2y + 4y(x^2 + y^2)$.

→ Question type → first move

Asked for...	Do this
plane through 3 points	$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$, $d = \mathbf{n} \cdot \mathbf{P}$, divide by d
plane $\supset$ line + point	$\mathbf{n} = \mathbf{v} \times \overrightarrow{P_0A}$
$\mathbf{Q} \perp \mathbf{V}$, $\mathbf{U} - \mathbf{Q} \parallel \mathbf{V}$	$\mathbf{Q} = \mathbf{U} - \text{proj}_{\mathbf{V}} \mathbf{U}$
tangent line to $\mathbf{r}(t)$ meets a plane	$\mathbf{r}(t_0) + s\mathbf{r}'(t_0)$, substitute, solve s
f at a point from f_x, f_y	integrate f_x , match f_y , constant from given value
df/dt from ∇f and rates	$\nabla f \cdot \langle x', y', z' \rangle$
constant from a directional derivative	normalise $\mathbf{u}$, solve $\nabla f \cdot \mathbf{u} = \text{value}$
tangent plane to $F = c$	$\nabla F(P_0) \cdot (\mathbf{x} - P_0) = 0$
type of critical point, only f_x, f_y given	differentiate once more, D -test
all critical points	solve the simpler equation first, substitute
best fit $y = ag(x) + bh(x)$	normal equations (tabulate)
max/min on a curve/surface	Lagrange; evaluate f at every candidate
max/min on a closed region	interior + each edge + corners

! Pitfalls that cost marks

- Normalise the direction before $D_{\mathbf{u}}f$. D -test: the sign of f_{xx} matters only when $D > 0$.
- Lagrange: never divide by something that may be 0; treat $\lambda = 0$ and zero variables as cases; an unbounded constraint may have no max.
- Speed is $|\mathbf{v}|$: keep $|\sin(t/2)|$, $|t|$ and split intervals. **Collide** (same t) $\neq$ **paths meet** (t, s).
- Rename the parameter of the second line ($t \rightarrow s$) before equating. Check the given point satisfies the surface.
- Answer format: exact fractions in lowest terms; " $ax + by + cz = 1$ "; $\langle a, b, c \rangle$ vs $ai + bj + ck$ as asked.